Finally…The Truth with Michael Shermer

Podcast: Startalk Podcast

AI Executive Summary

1. Core Thesis The episode argues that while human cognition is inherently flawed and prone to various cognitive biases, objective truth can be approached through a disciplined framework of scientific skepticism and fallibilism. The discussion counters the "post-truth" narrative—which claims all truths are merely subjective narratives—by asserting that science provides a method for reaching "provisional assent" through external validation and self-correction. Ultimately, the episode posits that the strength of the human intellect lies not in claiming absolute certainty, but in our ability to update our beliefs when presented with superior evidence.

2. Key Arguments & Data * **Definition of Objective Truth:** The guests define objective truth not as an immutable, final state, but as something confirmed to such a degree that it is reasonable to offer "procial assent." This introduces the concept of *fallibilism*: the recognition that we could be wrong and must remain open to new data. * **The Failure of Post-Modernism:** The authors argue against the postmodernist view that scientific truth is just another narrative. They contend that while all narratives exist, scientific narratives are unique because they are subject to empirical testing and external validation, unlike purely subjective or "internal" truths. * **Cognitive Biases & Evolutionary Drivers:** * **Motivated Reasoning:** The tendency to seek out information that confirms pre-existing desires or beliefs rather than seeking the actual truth. * **My-Side Bias/Tribalism:** The use of belief as a "servant of belonging," where individuals adopt specific stances (political, religious, or social) to signal loyalty to their "tribe." * **Patternicity:** An evolutionary survival mechanism where the brain finds meaningful patterns in random noise (e.g., mistaking wind for a predator). This leads to the formation of conspiracy theories. * **Hindsight Bias:** The tendency to perceive past events as having been predictable and inevitable, despite their inherent randomness. * **The "Dragon in the Garage" Metaphor:** Used to illustrate the concept of unfalsifiability. If a claim is constructed to be invisible, intangible, or undetectable by any scientific instrument (like a levitating, cold-fire-breathing dragon), it lacks empirical value because there is no way to prove it wrong. * **The Crisis of Trust in Institutions:** The decline of public trust

in scientific and medical institutions is attributed to the failure of these bodies to communicate uncertainty. Using phrases like "the best available evidence" can inadvertently erode trust if not coupled with a clear explanation of the probabilistic nature of science. * **The Impact of Social Media:** Social media acts as an "outrage engine," accelerating the spread of misinformation and making it difficult for individuals to distinguish between high-resolution, verified facts (e.g., news of a spy balloon) and grainy, unverified claims (e.g., UAP videos). * **The Role of Religion/Literature:** The guests suggest viewing religious texts through a "literary lens" rather than an empirical one, treating them as parables or moral guides rather than literal scientific claims.

3. Actionable Takeaways * Adopt a Fallibilist Mindset: Actively assume that your current beliefs could be incorrect. Periodically ask others to "double-check your calculations" or look for flaws in your logic to prevent self-deception. * **Distinguish Anecdote from Data:** Recognize that personal experiences (e.g., the benefits of a specific treatment or a psychedelic experience) are "internal truths" and are not a substitute for controlled, large-scale epidemiological studies. * **Practice Active Skepticism over Contrarianism:** Do not simply doubt everything for the sake of being a contrarian. Instead, use skepticism as a tool to move from a state of doubt toward a state where evidence is so overwhelming that you no longer need to doubt. * **Audit Your Information Sources for "Motivated Reasoning":** When consuming news or research, ask yourself: "Am I seeking this information because it is true, or because I *want* it to be true to support my side?" * **Apply the Correct Lens:** Use different evaluative frameworks for different subjects. Approach scientific claims with empirical scrutiny and historical or religious texts through a lens of cultural and symbolic meaning.

Full Transcript

So, Gary, are you objectively true? Maybe. But then again, there's your truth, my truth, and Chuck's truth. And I know for a fact that mine's a lie. Coming up, the Star Talk Special Edition. All about truth. Is it still there? Coming up. Welcome to Star Talk. Your place in the universe where science and pop culture collide. Star Talk begins right now. This is Star Talk, special edition, which means we got Gary O'Reilly right here next to me. Hi Neil. Gary. Yes. I love having you. I love being here. Yeah. Chuck, how you doing, man? Doing great. So, you cooked up an episode, you and your producers cooked up an episode today that, oh my gosh, this is an eternal subject. And take us in. The truth. Is it out there? Yes.

So apparently he's trust, but do you trust the teller of the truth? Everyone trusts me. I'm the most truthy person that you'll ever meet. I make up the truth. I'm so good at it. Is it noise? Is it biased? Is it fake news? Are they alternative facts? But then stop and think for a second and ask, do we actually examine the truth differently for, say, science? or then for history or politics, even religion. Well, how about UAPs and aliens? Yeah, might talk about them as well. What if there was a toolbox of sorts that could help you build a skill set that could get you closer to what you might call the true, true? It's not the fake truth, the true truth. So Neil, I think let's bring in our guest. All right. Well, I'm delighted and honored to bring in someone who I can claim as a friend for decades. It is the one and only Michael Shermer. Michael, welcome back to Star Talk. Nice to see you all. We don't go back to a previous life. Maybe, maybe. We'll work on this a different episode. And welcome to my office here at the Hidden Planetarium. And you're author of a dozen books, each one trying to get people to think straight in a different way. You are a founder of the Skeptic Society, is that correct? And there's a magazine called The skeptic. Yes sensibly titled. That was a focus group. I like to call it fakeness. So and you are also host. You're also host of the Michael Shermer show. It's your own podcast. And how did you come up with that name for it? It was original, huh? No, we're good. And your latest book. Oh my gosh, because this this.

could have almost written itself in the time and the place that this needs to appear. It's just called truth. Man, that is an ambitious title, I gotta say. No, no, no, here's the problem. The real problem is that this book is even necessary. Well... Who would have thought? You know, yeah. I mean, in this day and age, it's far more necessary, though. That's what I'm saying. I'm just saying. I'm sad that you have to write it as well. Well, thank you. Yes. Well, that's the problem. And so I start with the idea that there is no truth and we're living in a post-truth world, which the moment somebody opens their mouth and says that it can't be true because they're making an argument.

Which is searching for truth. Wow! Hey! Can we start in the top of the pool first? Kid pool first! I mean, my man, it's like he's standing in the square in ancient Greece, just like... I'm looking at these lifeguards with this. Man! Okay, we'll kick... Can we unpack what you just said? Yeah, man. Well, so there's people, you know, fake news, alternative facts or postmodernism. There is no truth. My truth is as good as your truth. Scientific truths are just another narrative, just another story like anybody else's story. So these are what people say. Nothing's special. This is what people say, right? So we counter that. These are the so-called science wars. We counter that by saying, actually there are objective truths and we can give that them. Here are the methods we have. That's chapters two, three and four of how to get there.

All right, so, so wait, I'm sorry, before you go further, can you define objective truth then? Because if we don't do that right now, then everything else we say is kind of like do that right now. It's really kind of flimsy. Something confirmed to such an extent, it would be reasonable to offer our provisional assent. Something confirmed to such a degree that it would be reasonable to offer Our provisional ascent. Our provisional ascent. No, that's the key word in there. It could be wrong. I could be wrong. That works on a lot of lifted microphones. I know, so I start with the premise that whether there's a God or not, I know for sure it's not me and it's not you or you or you, sorry. That's right, we know who it is. I'm sorry. Go ahead. How do you know it's not any of us? Well, because we're all, so it starts with fallibilism. We're all could be wrong about anything. So we start with the idea that I may have my personal truths, but I might be wrong. So I need to

ask you to double check for me. If I think I believe X, can you look and double check for me to see if I'm hallucinating? How many people do that? Almost nobody, which is why I wrote the book, right? Well, you scientists do. Yeah, we do. But you're trained to do that. Anytime we come up with an idea, I say, can you check this? Did I make an error? Can you double check my calculations? You actually go get your haters. Yeah, that's called the peer review. Ideally, you get your haters and they come in and they go, yeah, this is bullshit.

So you remember Carl Sagan's dragon in my garage story? I heard it, but I was never able to retell it. Okay, I'll retell it. If you can please. There's a dragon in my garage, would you like to see it? Oh my God, that'd be so cool. So I take you to the garage, I open it up, you look in, there's some bicycles and paint cans and a ladder. You don't see any dragon. I go, well, it's an invisible dragon. You go, oh, okay. How about we sprinkle some flour on the floor and we'll catch the footprints of the dragon walking around. Well, you see this is a invisible levitating dragon. It hovers above the ground about a foot. All right, how about we get one of those thermometers they used in COVID where you pointed at people's forehead and get the temperature? We'll see the outline of the dragon walking around in there. No, this is a cold-blooded dragon.

How about the fire? But just to be clear, when you're cold-blooded, you are the temperature of your environment. So they don't necessarily have cold blood. They just don't maintain... They don't create their own warmth. They wouldn't show up on an infrared camera. That's right. Or the fire. It's cold fire. What's the difference between an invisible, levitating, cold-blooded, cold-fire-breathing dragon and no dragon at all? and you can't say, well, I feel the presence of the dragon when I'm in the room, or I feel the love of the dragon. That's all fine for you personally, but at some point for us to believe there's actually a dragon, we gotta have some evidence. And that's science. By the way, Michael, I just wanna go on record and say, if an old white man comes to me and say, there's a dragon in my garage, would you like to see it? I'm calling the police. That's how you respond to it. Okay, now suppose, just walk while we're there, okay? Suppose you are so charismatic.

You were so lovable. You were so articulate. Sure. That anyone you speak to of this dragon ends up feeling the presence of the dragon just the way you do. Out of the power of suggestion. Why doesn't that then constitute evidence? Because that's how you get cults. Yes. All the people in the cult feel what the leader of the cult feels. Why doesn't that constitute a certain kind of evidence? So for science, we need external validation of your internal state. So maybe you do feel this, fine, that's your truth, but for me to believe it, I need to have some external validation. So if I take ayahuasca and I realize, and I tell you, oh my God, there's this all of the spirit world. I saw the entities, they were floating out there. Here's the color. I felt their love and so on. And you go, can show me the evidence. And I go, here, you take the ayahuasca and you'll see it too. That's still in your brain, you're still at internal truth. We need to have some, a measure of it. Now, suppose that, and I'm playing devil's advocate here, because what you just, I've actually experienced that. With ayahuasca. With ayahuasca. You did? Oh my God, it was amazing. Right. And I always- I keep on it now. Did you feel like there really is this other place that you can- It's more than I'm feeling, and that's why people are so incredibly passionate about it. And just to be clear, you took it under very controlled setting. Very controlled in a place where- Yes, they have medical staff on board. And if I come back and go, Neil, I was wrong. There is this other spirit world. You will not come back and say that. Now you might say that there's a possibility that there's a spirit realm and that somehow in the ultra consciousness, we're able to tap into it, but we can't do it otherwise. You might say that, but the truth is everything that the entire time that I was going through, and you're not high, that's the misconception. You're not high, okay?

as you're in this altered state, the only thing I kept saying as I was looking at beings who were in front of me, it wasn't like a hallucination, a hallucination, whatever, I'm there there, to me they're there, but I knew they weren't there. And I kept saying these are artifacts of my mind making. I kept telling myself that, however, what my mind, through this experience, revealed to me was so deep and so useful that it felt like this couldn't have come from me. But I know for a fact that it had to come from me, okay? Because- Well, you're rationalizing it. Well, I'm not even really, I am rationalizing, you're absolutely right, I'm rationalizing. But someone less rational than you, it's a whole other being- Oh my goodness. You're all in,

they're all in. By the way, it still exists as that for me as well, because it's so real. Like I said, it's more than a feeling, it's a connection that feels completely external. But guess what? I know for a fact that it was coming from my brain. Okay, so Michael, getting back to the dragon in the garage, with a charismatic cult leader, one by one people are drawn in, and for them, the evidence is that they feel it. So at some point you need someone who was not swayed by the charisma of the leader, who then would be counted as an outside reference. That's why cult leaders don't want you to engage with people on the outside. That's right. Yeah, okay. Or you need a device. Or you need a device. Well, if I could just say something about your experience. So in my model of truth, there is a realm of personal truths that...

are true for you and do make a difference in your life. So when somebody says, I found Jesus or God or whatever, and it changed my life, I'm not saying you should believe, I'm not saying I can prove it. I'm just saying this is what worked for me. I'm fine with that. I think that's wonderful. Especially if it changes for the better. Hopefully. Is what Chuck was describing with his experience, what people are calling truthiness, the truth you want to believe? Well, a little bit, but that was Colbert's riff on more political. kind of trues. I think this is more in trues. But is that the trues that you personally believe and the one that you want to believe? Yes, yes, that's right. Yeah, it's an extra step there where I would really like this to be true. Now I'm going to find the evidence for it. That's not helpful. This is slightly different. Again, I think if there's experiences you have that change or people have near-death experiences, did they really go to heaven?

Probably not, but it changed their life. Now they've, you know, they're doing this and that instead of what they used to do. Okay. Yeah. And I will say the root and I'm going to get off the ayahuasca thing because that's not what we're about. But the biggest benefit to me was I was on several neuro transmitter medications and I was able to come off all of them under the supervision of my psychiatrist who was like, Hey man, I don't know what to say. like we don't have these answers in neuroscience right now, but clearly something happened and boom, and I haven't taken them since. So that's- I would like scientists to be more open-minded about that sort of thing. They tend to be pretty dogmatically skeptical of that. I think we need to keep the

door open for these kinds of experiences just in case. But part of that skepticism is because if you were a betting person, most of such claims would not pan out. And so if a scientist has limited time, you're gonna say, well investigate this, because it surely works, but you're just in a line of people making exactly the same suggestion of the time of the scientist. So go back to the dragon scenario. Yeah. And you've got the, I suppose, deception of a false narrative. But are we programmed to be self-deceiving in the way that we will...

Absorb a truth that we know may or may not be that's because you started saying there's the truth You want to be so and then you look for evidence in support of it a good scientist will certainly want their hypothesis to be correct. But a good scientist will try to find ways to show that it's not. That's right. There's two very different paths. If you don't find the flaws in your research, somebody else will. Somebody else will. With great glee and they'll publish it. Right. So the whole development of science is to get around those issues. Our brains are more like lawyers than scientists. We want to win the case for our side, not find out what's actually true. So that's what we have to work against.

This is Star Talk with Neil deGrasse Tyson. It's how you structure it and bring it. Is there a standard issue? Well, there's tools we use, you know, correlation and randomized controlled trials and these sorts of things, statistical tests, that science has developed over the last century and a half or so to see... But how about as an individual rather than as a scientist? We're more subject to this motivated reasoning. I want this to be true. Now I'm gonna just collect all the evidence that confirms it, ignore the evidence that disconfirms it. Everybody does it. Everybody does it. Motivated reasoning. I like that phrase. Yeah, we're motivated to find what we already want to be true. Find evidence for it. And that's the problem. At some point, can you really fault people for that? I mean, we're human, right? We're subject to the whims of our emotions.

And also in our desires we have to trust our institutions and Well because who can fact well you don't have to but who can fact check everything? I mean, I never saw the Chinese spy balloon with my own eyes, but I accept the high resolution photograph. There was the jet shooting it down. It was

on all the news. The president, the secretary of defense, they all said it's real. By contrast, the UFO videos are always grainy and blurry. You can't quite make it out. No one will confirm it. It's all like, oh, it's classified. We can't show you. That's very different. Two different examples of news reports. One has a level of evidence that doesn't leave you skeptical of the claim because the images are that's why we were pretty high resolution. Yeah, you can see you know So what do you do with the I'll call it the trust of the masses? Where so many people believe something, but the thing that they believe is still wrong Yeah, so my sorry Chuck. Yeah, that's my side. Yeah, exactly. What is my side bias? My religion is right. My political party is right if you want to go into the Trump The Trump spirit comes upon him. This is the first time anybody's ever killed Trump. We don't summon Trump. Trump appears on his own. He's a channeler. So again, I mean, the levels of trust the public has in scientific institutions, medical institutions, and the academy, universities, has plummeted to an all-time low post-COVID. This is a problem.

So that's one of the reasons I wrote this book was like, hey, these institutions actually work pretty well. They usually get it right. Sometimes they're wrong. I would have preferred our institutions to couch their recommendations in more probabilistic. Right now this is what we think about masks or vaccines or whatever, social distancing. 50-50, we're gonna go with this. Maybe next week we'll change our mind. They could have done that. That was the worst thing they did is the fact that the best available evidence. No one ever used those terms, but it's hard to engender trust when you preface everything with the best available evidence up until now, and then people go, well, what's that supposed to mean? I think there's some hope there because when we grew up, I think we're in the same decadal age, polling results were not given with uncertainties and now they are.

Okay, they just gave a number and that was like the number right now. It's plus or minus three points, right? So I thought was an important achievement I think you'll hear people say things like my priors are this and I changed my upgraded by priors and my credence is now that that's a whole Bayesian area of statistics that people have no idea what they're actually saying But they've inculcated it from just pop culture. Mm-hmm.

And this is a good thing like it is I'm gonna it's okay if I changed my mind. Yeah, yes, yes, absolutely and I think that's very important too, to give people the leeway to say, it's okay if I'm wrong. The best thing I like about science, it is hands down the number one thing since I've been working with Neil for all these years. The thing that I appreciate the most is information that has changed when better information has come along.

And nobody, the world, the science world is not set on fire. They're like, yeah, okay, well, we got better information. Now, one proviso on that is that just because that idea was wrong, and that one was wrong, my idea is surely right, even though you think I'm wrong. I'm one of the heritage, I'm Galileo. Right. I'm right, yeah, and everybody else is right. Does AI help or hurt this exercise? It's a tool, I don't know, we don't know yet. It's a useful tool, which is gonna- In the search for truth. Yes, yeah, absolutely. But you gotta check the sites, the sources, they cite. If they provided the gifts. Some of the sources, they're hallucinating, they're not real. If you click on the link, it's like, that article doesn't actually exist. When we spoke to Jeffrey Hinton, he said, some AI does have overconfidence. Overconfidence. He said that, the hallucinations that we all know and laugh at. He doesn't call them hallucinations.

Confabulations. Confabulations, that's what he said. That's good because we do that, humans do that. That's why he calls it that. We have fragmented memories of things and we fill it in so that the narrative can make sense to us. Reagan famously remembered things from movies he was in that he thought actually happened. It's like, no, that was a movie you did in fifties. Oh yeah. So if we go back to the my side bias where people are believing things even though they know it's probably not right. Is that true? Oh, sorry. That's very meta. And I came across the phrase, and I'm not sure if it's yours if it is, I appreciate this, a servant of belonging. That's not my phrase, but I like it. What does it mean? Well, so when people publicly signal, like the example I use in the book is, you know, asking undergraduates, do you accept Darwin's theory of evolution? They go, oh yeah, of course, yeah. Explain it.

Well, they can't, you know, the draft stretches its neck and the baby drafts have long, no, that's not it at all. So they're publicly signaling, yeah, the

scientists tell us this is true. So it probably is. And so a lot of conspiracy theory proclamations, yeah, I think they're queuing on Israel or the pizza gate thing. I mean, if you really think about it, this cannot possibly be true. But what they're signaling is my side says, this is a conspiracy theory. I do know there's bad things that happen. Those Democrats, they're up to no good. I don't know if Hillary actually did this at this pizza gate. But she did something. She did something. That's it. That woman's guilty. Find her. Find her why she's guilty. I call it the past interference syndrome. If you're rooting for the team that the call was made against, it's not past interference. That's right. If you're rooting for the team that the call was made for, it's definitely past interference. That's right. And you actually say it that way. That's right. You know. It's back up a moment.

just to get the vocabulary established. Will you refer to Bayesian? reasoning, could you just say something about that? So that's Reverend Thomas Bay's investment. It's kind of statistical methods of when you should change your mind when the evidence changes. So you start with here's what I think is true based on the following points of evidence. Now, if those change, I may change my credence, that is my belief, my confidence that this thing is true. And it's just an influx constantly changing. This allows you to not be so committed to the belief that I have to defend it like it's my territory and it's under attack. Rather it's maybe true, maybe it's false. I don't know. This is my confidence level now. Let's go there. So it's a more fluid. It's more fluid. It's a more fluid tracking of. So is there a point where truth is fixed and fixed only? It can never be changed. Not in science. I mean, if it's a religious truth, this is what we believe. All right. So I'm trying this out in the book that a lot of religious truths are actually more like literary truths. They're just stories.

that have deeper meaning behind them, like great literature, you know, a Jane Austen novel or a Dostoevsky novel. Is there really a Middle Earth somewhere on the planet from J.R.O. Tokens? Don't ruin my dreams. Maybe it's in New Zealand. Sorry, sorry. Exactly. You're making a really good point where we don't approach fiction declaring that it's false. Right, of course. That's not the operative way to think about it. We approach fiction as a source of lessons, parables, ways of living. Moral compass can come out of it. And so that's why I've never liked it when people say, is this

factor, is it fiction? Because that equates fiction with being false. And I'd rather fictional literature not be thought of as true or false. Same with mythologies.

Is there a lesson in there? Now, pick up where you were going. Yeah, so like the story of Jonah and the whale are the great fish. Did he really live for three days in the belly of a great fish who'd then spit him out at this other land where he goes to deliver the message, the moral message to these other people, okay? you're missing the point of that story. It's not, you know, how do you live inside a fish for three days? You know, the three days is very symbolic, okay? And most Christians when I, you know, of course I don't know, that's probably not real, but you know, there's a deeper truth there, okay? How about the resurrection? Maybe that didn't actually literally happen, but it's a story about redemption, forgetting this, starting over and they go, hang on, time out. Time out, time out. Red card, yeah, red card. You've got Moses in the parting of the Red Sea. You've got Moses in the parting of the Red Sea.

You can't really literally say it has to happen. Well, I've heard people have sent this article as a skeptic. Oh, that was caused by an earthquake or this tsunami or this volcano. Maybe it didn't happen at all, and it's just a story. And in the case of Jesus, there's people that think, well, this is called the swoon theory that he wasn't actually dead on the cross. He just painted, he swooned. And then they put him in the cold cave that preserved him for a couple of days and he woke up. And then he married, he went with Mary Magdalene. And if you want to go full Dan Brown, they moved to France, had kids. Full Dan Brown. I get the reference. That's funny.

And the Illuminati's protecting their lone descendants today. So again, most Christians go, no, no, no, it has to literally happen. Now, why do they insist it literally happen? I think part of the reason is because we live in the age of science where your core beliefs should have some foundation in evidence. In the age of science. Yes, I just think, okay. So they're always swapping out the lenses for different subject matters and the way that we are. we would look at religion and biblical stories through one lens, and if you take politics or something else, and we talked about UAPs at the beginning of the show, and we see them in a totally, through a totally different lens, maybe more

skeptical, maybe less skeptical. Well, this is my point. Use the right lens to look at this thing. Read biblical stories as literature, rather than as empirical claims. Absolutely. But, you know, unfortunately, he's going to hell. Just saying.

We started our little chat here about post-truth era. Are we? Are we talking towards it? Please say no to that. Yeah. Please. Well, that was the opening. So here's before you even say anything. Yeah. I'm really embarrassed to admit this now. What the hell is post-truth mean? What does that mean? Pay attention. That's why we're asking a man. This is what I'm asking. What does that even mean? He's got a book with this title, see? So shut the hell up and let the man talk. Okay, go. Well, what it's supposed to mean from the kind of post-modernism and all that stuff in the 80s and 90s is that The old model was that there is an objective reality and truth that we can discover through science. The new reality is that science is just another narrative story that people tell and it's no better or different than any other stories. So the aboriginal Australian stories about the origins of life and so on is just as good as your scientists. And so what we're pushing back against that is saying, no, they're not equal. And so that would be post-truth that there's no truth to be found.

But when you actually read postmodernist, they're making an argument and they're going, here's our evidence for it. It's like, that's not my post-truth, you're making an argument. So you're on the page with me, like, all right, let's see what you got. So then on the back of that, do you get the feeling that there are segments of society, whether it be Eastern, West or whatever, that have just given up on the truth and have more or less just switched off? Well, some people have, yes. I mean, that was, remember, Steve Bannon's famous line to Trump was just flood the system of shit, and then we'll tell them what's true, because they won't know what to think. That's right. He actually said that? Yeah, he did. Yeah, he did. Oh, yeah. Yeah. This was in, like, 2015. He's a great guy. I gotta tell you that Steve Bannon, he's something special. I was on a real time with him on one episode. Steve Bannon? Yeah. Yeah. Yeah.

He seemed pretty okay at the time. See, that's the problem. Isn't that the trick of the devil? That's the problem. He seemed like a regular guy.

Exactly. He seemed, I'm just saying. That's the disturbing part. So, can you give me a short list of the kind of biases we should look out for through ourselves? And how do we know they're not in your book? Well, you have to read the book. Check the evidence there. Well, so obviously the big one motivated reasoning. Again, we're all just designed to look for and find evidence for what we want to be true. He said we're designed. Well, designed by evolution. So there's one theory that reason evolved as not to find vertical truth, what's actually out there, but to win our arguments for our side. That's one case. Now, you know, we have to have some capacity to understand what the world is actually like in order to survive. So there's some probably a balance there, but the idea is that we're gonna win the argument for our tribe. That's the my side bias. The hindsight bias, you know, after the fact, you look back at how things unfold and you go, oh, it obviously had to be what that way. My life had to come about that way.

what happened. But in fact, most of it's just random. And we're not good at detecting randomness in your business. The stars in the sky are clustered. That's what randomness looks like. My favorite story from Steve Jobs is when they released the iPod shuffle feature that randomly played your songs up from your playlist. And people complained and said, certain songs come up more than others. It's not random. It's like, no, that is what randomness looked like. So they had to program it to be less random so people would feel it's random. You program it so that it hits. Every song before it then right we does it, but that's not random That's not random at all. Yeah, that's like throwing the die and getting all the numbers We're cancer clusters. Oh look there's this community here We got a look to see if there's an industrial waste stuff there at some company So they find the cancer cluster first. Yes, then they look for what could cause it but there's other Clusters where there's no cancer at all in this community and no one goes in there to see let's see what they're doing, right?

Because they ignore those misses right so That's the one of things that you know epidemiology. They have these tools like we have a cluster of no cancer. Yeah. Yeah, I mean There's like well these people are healthy That's not even a name for it is not even a name. It's not gonna make the news people are healthy No interest. And if they're living near high power lines, no one's gonna see the high power lines made them healthy. So what kind of

bias is that? Well, that's just the, well, I call that patternicity, tendency to find meaningful patterns and random noise. Most of it is just random. Like Jesus and a piece of toast. But we're really good at that. But for reasons that I think are plausible, with the lion in the bushes.

Yeah, so you hear a rustle in the grass as a dangerous predator, is it just the wind? So if you think that the rustle in the grass is a dangerous predator and it turns out it's just the wind, That's just a false positive, no big deal, you just move aside. But if you make the other kind of error in which you assume the rustle in the grass is just the wind, it's a dangerous predator, you get a Darwin award for taking yourself out of the gene pool early by making that mistake. So we tend to assume the worst, we tend to assume the pattern we think is real is really Israel just in case. Same thing with conspiracy theories, most of them are not true, but some of them might be. Maybe it's better to err on the side that they really are after me.

even if they're not after me, they might be. So I better be cautious just in case. It's funny when you say that because I have talked, I do a lot of work in climate. And one of the things I say is everybody loves conspiracy theories except the one conspiracy theory that's in front of everyone. And that is a small group of companies got together and decided that at all costs possible, they were gonna make sure that we dig shit out, I'm sorry, that we dig stuff out of the ground and burn it full well-knowing since 1950, whatever, that it would destroy our planet. Now that is a conspiracy! But you say that to people that are like, Duke!

Get the hell out of here. Well, same thing with tobacco, big tobacco. They knew the deal for a long time. That's a conspiracy. But these conspiracies, they're not sexy. They don't have aliens. And so nobody cares. That's right. I mean, the Illuminati's, these 12 guys in London running the world. Okay, but there are 12 CEOs in tobacco that, you know, they had meetings. We have their memo. And they knew the deal. And you know, millions of people died, but yet somehow that's not a conspiracy. They're the ones that, you had that phrase, you know, the jury is not in or the science is not settled. We have to wait. We have to wait. That's where they came from. This notion of settled science, the people I've spoken with who would invoke that term, the way they bring it to the table is there are these two

research papers that are opposite the other hundred. And so therefore the science is not settled. Not settled.

without realizing that there's always going to be outlier research papers. And so at some point... Especially when they're paid for by the people who don't want a different answer. So tell me about the notion of consensus here. Because when you look at... I think if you look up the definition, it refers to opinion. Consensus of opinions. And in science, when we speak of consensus, it's the agreement of experiments and observations. So how has that word worked for you when you've been invoked in? Well, in science, yeah, we have to... be careful to find that. I mean, it's a consensus of the experts who have been finding each other for decades about whether this hypothesis or theory is true or not. By the time we consume it, they've already hashed out all the problems. This is what we think is probably true.

It could be wrong, but that's what 97% of climate scientists say X. What about the 3%? Maybe they're right. Maybe, but probably not. Yeah, Mr. Exxon and Mr. Shell. Mr. BP, Mr. Exxon and Mr. Shell. I mean, Neil, you must get these papers. You know, I've been working in my garage here and the, you know, Einstein was wrong and Newton was wrong and Hawking was wrong. And I got the theory of, I'll share the Nobel Prize if you help me with the math. I mean, you get these. Okay. So how do you know that guy's not right? I mean, it could be maybe all these others, probably not, right? Cause usually the outsiders, the heretics and so on, the would be Galileo's, they're not would be, they're not Galileo. They're just wrong. So if we take a holistic view, and go in and look at all the different components of a truth, our truth, my truth, your truth. What value is truth to modern society right now? I mean, we can't run a society in which there's not some agreed upon reality. Like money. I mean, it's fiat money. It's just ink on paper. It's not valued at all by itself. But with a structure behind it that says we value it at this level and we will back it up.

Okay. And so we all have to agree to that though. And then once we do that, that becomes the reality even though underneath it, those are more political. That means people have the power to create a truth. In that sense, yes. So political truths are a little different than others. In what way? Well, so in a way, a democratic election is a kind of an experiment. What's the right

percentage of immigration a country should have? Zero. There's no right number. Unless, of course, they're from South Africa. Or my wife. Or my head of the FBI. Exactly. Or all with parents who are immigrants. Wow, look at that. Yeah. So we run an election. That's an experiment. We go, OK, these people are now in power. They're going to set it here. And we'll see how that goes.

And then maybe the public goes, yeah, we didn't like that. We're gonna bring in the other team and they're gonna do something different and we'll see how that goes. Yeah, try to dare you. Well, so, but when you talk about, when you look at it that way, you bought up money, okay? It's not just the trust that it's valuable. It also behaves as if it were valuable, which is why we then continue to put more trust. So for instance, if we had runaway inflation, our dollar would become worthless and we would stop trusting it. Therefore it would not be of value. So the levers that we are pulling to keep the dollar as a viable world currency is the thing that gives us trust. Therefore it really does have value though. That's right, yep.

I mean, that's where the government implemented the, you know, FDIC and that, you know, we're guaranteeing a certain level of money in the bank, even if you think it's not really there, you don't have to have a run on the bank. That's all based on exactly what you just said. A bank run is people just don't trust that the money is in the bank. Gotcha. And it's actually not in the bank, right? It's out on loan. That's who you're right. So it's too many people go to the bank. It's in your house, and it's in your house, and it's in your house. Carry out all the money out. Exactly right. It's a wonderful life. I'll be experiencing more disinterest in the truth. Well, I think there's an incentive to say that. I think people actually still want to know. Are they getting steered towards this? By political groups, religious groups, whatever. Yeah, that's happening. Probably more than before.

but it's hard to say. It could just be elevated because of social media. It just becomes so prominent, so quickly in real time, you can see what's happening on X or whatever, where people's attitudes are. I doubt this or I doubt that. I mean, just like the ice shooting of Renee Good, right? So you have the exact same video clip. You go Fox News, you can clearly see here, she's trying to run over the agent. You go over to CNN, you can clearly see

here, she's trying to escape the agent and not cause a problem. We're looking at the same video. So even the video evidence is not going to be enough. Not enough, yeah. Are we failing to adapt to the speed of social media? Are we going to struggle even more with social media? I don't know. The whole thing collapse is so new. I don't know what's going to happen. Yeah, because it's feeding you, you know, it's an outrage engine. But let me ask a different question here. You are doing the bidding of science.

And scientists, yet you yourself are not a scientist. What is your disappointment? Wait a minute, what? No, I'm joking. Why am I a historian of science? Right, that's important. That's your background, the history of science. So my question is, why aren't more scientists doing exactly what you're doing? Why are you doing the bidding of scientists? Well, they're too busy working in labs actually doing real research. Because what you say is exactly what's going on, but the scientific community doesn't articulate it. We're not trained to articulate it in our graduate. And you're not built for it. I mean, let's be honest. I'm often accused of defending the orthodoxy. I thought you're a skeptic. Why aren't you skeptical of whatever their pet peeve is about science?

Well, because of the consensus thing, I trust the experts. Maybe they're wrong. Maybe you're right with your theory about vaccines or the Big Bang or whatever, but probably not. And so I'm really just, what I'm expressing is the system works pretty well most of the time. And even when there's a mistake made and there's a correction, all that theory turned out not to be right or this or that, expanding, accelerating, expanding universe, all that stuff. But that wasn't scientists just being wrong. It was scientists correcting the updating, upgrading our and so on, and finding new evidence, that's how it works. So even the worst mistakes in the history of science that, like creationists point out, built down man, they got that one wrong, and they got that fossil wrong. Yeah, but the mistakes were made, discovered by scientists. Right, right. But it wasn't the journalists finding the mistakes. That's right, yeah. The scientists found their own mistakes. And even after the replication. Because the system is built to do that. That's right. So no, unpack our understanding of the word skeptic, because I've heard people saying, I'm just being skeptical of what the scientists were telling me. Isn't

that a healthy posture? At what point do you say, because one of the skeptics organizations, if not yours, I don't remember which, had a button that button just said, I doubt it. Oh yeah, that was Joe Rogan. No, and I'm just saying, no.

Being a skeptic means you know at which point you no longer need to doubt it. So where does that training come in? Because a lot of the people who are, quote, skeptical of vaccines, like you said, skeptical of masks, skeptical of any emergent scientific objective truth, They're just being skeptical. How could you fault them for that? How do you take us out of that? Well, so at the beginning of a- I'm just asking questions. Just asking questions. I know that's a very dangerous, because they're usually not just asking questions. Absolutely. They're moving us down a pathway. So we start, in size, we start with a null hypothesis. Your hypothesis is probably not true. Now, go ahead and give us your best evidence. I think Iver Mectin will cure the issues with COVID-19. That's nice. Okay. Do you have any evidence for this? Well, my aunt tried it and she got better. Okay. Sorry, in medicine.

then an anecdote is not data. We have to have controlled trials, epidemiological studies, and so on and so forth. The plural of anecdote is not data. That's right. I think Bigfoot's real. Okay, it could be. Maybe there's another bipedal primate running around out there in Canada or the Himalayas. Show us the body. And if they do, okay, then it's real. But if not, then I'm gonna remain skeptical and tell I see that evidence, that's completely reasonable. All people should be scientists in that. So there's nothing special about science and skepticism in that regard. We should all be skeptical to start, but then be willing to say, okay, I think that one's probably true. So there's some people who believe they are nobly skeptic, who don't know that boundary between when the skepticism is healthy and when it's just wrong to be skeptical. That's right. Yeah, there are people that just sort of, Contrarian by nature. I'm just going to doubt everything just for the hell of it. There's the bottom. I doubt it. I don't believe anything anymore and skepticism. I mean, the line's quite blurred right now.

Back in the 90s when post-modernism was big and there is no real truth. I said, so the claim that the Holocaust never happened is as good as your

claim that it is? Well, no, okay, I'm not saying that. I'm like, oh, okay, so there is a way to find out whether something's true or not. Yeah, I guess so. We catch us up on this, because I don't want it to go unresolved, because the 90s was 30 years ago, okay? What? That can't be. We're just making up math now. 30 years ago. 30 years before the 90s was the 1960s. So this postmodern movement was led in academia by the liberal arts community. Could you just remind us about that? Well, the humanities, the literature, the lit crit and all that stuff, deconstructing the text. What did the author really mean when they said this? And we can't know because they're gone, whatever. So it kind of came out of that. And then so we're examining narratives. Well, science is a narrative. It's a story. Then it became, it's just a story. It's just another story.

There's nothing special about it. No better or worse than anybody else's story in any other profession. And then it got wrapped up in the whole, well, it's, you know, it's Western industrialized, capital is white, you know, male. Okay, but that was pretty strongly seated in academia. When did that spill out into the public? Well, again, probably in the 90s, and just like any kind of claim that anybody makes, we should doubt it just because it came from a white guy or it came from an academy where this capitalist were running it or whatever, funding it, that kind of thing. Okay, what is the health of that movement today? Well, it's still around, but I think it's been challenged. I mean, to be fair on the right, there's also the kind of, fake news, alternative facts, you know, all this stuff. You can't trust the media. Oh, because the post-modernism was a leftist movement. That was a leftist movement. Yeah, yeah, yeah. So there's one on the right too. Yes. And again, one reason for the book is that we have to get back to trusting experts. Expertise does matter. Definitely. Even if they occasionally get it wrong, that's fine. Here's the deal. They may sometimes get it wrong.

But your dumbass always gets it wrong. You know, that should be on a poster. That's a poster. That's it. And Michael, there are people who say, I did my research. And they spent an hour reviewing YouTube videos. When a scientist says, I did my research. They mean something completely different. That's right. Yeah. So I like independent journalism. I think it's a good corrective for some of the errors of mainstream journalism, but most of those people don't have fact checkers or editors. So they also get it wrong

a lot. I just did an op-ed piece for the Washington Post. There was just 700 words excerpt from the book. Four people went through it. I mean, it took like four days and I had to like provide a link for every single sentence. And I was, to the point I was getting irritated and I'm like, actually, this is a good thing. There's people checking me. Exactly. Because I know when I'm just posted on some website that, you know, there's nobody fact checking it. Yeah. So let me ask you this. And I think I'm opening up a Pandora's box here right now, but I don't care. I gotta do it. You gotta care. It's just so important in this country right now, after almost officially eradicating or officially eradicating measles, we now are at a 35 year high for measles outbreak. And there is a direct correlation between the number of people who are not vaccinated and the increase in the cases of measles. What do you say to someone who is listening to the voices that says, Well, vaccines are like, they're harmful to your children. They're wrong. And there are clearly cases where people have died because they've been vaccinated and all the things that they put together, how do we talk to someone like that?

Well, you have to ask questions first. Where'd you hear that, you know, oh, on Tucker Carlson or, you know, Rogan or whatever. I was listening to RFKJ talk about, okay, so, but those sorts, you have to counter the false information with more accurate information. So, I mean, there are people that are harmed by vaccines. That does happen. You know, if you give a billion people something with a .0001% side effects, you're gonna get some, you know, enough to fill the evening news, right? So you have to counter it with data. This is the answer to what's the harm of people just being skeptical. or whatever, well, some people won't get vaccinated. And then they'll get measles and things like that. That's unfortunate. I mean, RFKJ, some of the stuff, you know, we should eat healthier. Yeah. No doi. Yeah. Just to be clear, the Liberals were OG anti-vaxxers. Yes, that's right. It only crossed over the fence to become purple when...

It became mixed in with the freedom. Can't have the government tell me. The right has never really been anti-pharma from my read of the news. No, because they're fine with pharmaceutical companies making money. But some liberals are like, I don't know, comfortable with big pharma making money on vaccines. So maybe the vaccines are better. No, no, no. I've got just a couple of minutes left here. Have you found anything other than

science that has access to objective truths? by accident. It's God's truth, so so. We took this long. There it is. We took this long for you to go there. Oh, go ahead. Well, I mean, they may get their by accident. I mean, some of the claims about religion, about how we should live our lives, turned out to be good. You know, you should treat others like you would want to be treated the golden rule. Okay, you could...

You can graft a game theory model with math on that. Yeah, that actually is true. Okay, they got that through experience trial and error over thousands of years. Okay. So yeah, some institutions like they can get it right without actually testing. You know, science is so new, right? We've been bumping around for thousands of years, getting some things right and some things wrong. But we're making progress. That's cool. Quickly or slowly? Slowly. Moral progress happens on a decadal level. It happens just slow enough you don't quite notice how it happened and then you look back to the 90s. And look, how are we done? This is such a thing as a moral truth. I think so, yes. So this is published by Johns Hopkins University Press. Wow. And how many of your books have been with him? Two, Conspiracy and this one. Okay. Yeah, they're good. They're really good.

the publisher. Yes, they fact check. They do a lot of fact checking and editing. Is this out now? Yeah, just came out. Okay. Yep. So what are you working on now? And how do we find you on social media? skeptic.com, michaelshrimmer.com, on Twitter, X, at michaelshrimmer. So Michael Shermer is the key. Yeah, that's the key term. And you post with some frequency. Yeah, yeah, not as often as you, maybe. Yeah, I don't depends what's going on. Do you have a TikTok account? I don't. I should get what I know. Not even on Instagram. I mean, I'm so old school. I'm older than you. Old man on the porch. I'm more on the porch. Yeah. Yeah. All right, Michael, take us out with some hope.

Oh, well, again, I think there is an objective truth to be found. I think we're able to find it provisionally and that our brains are smart enough to figure out what's really going on and understand reality. Even if it's error prone, we do it better than... Other species so we'll dig out of this. I think in my ultimate chapter is on at some point We could become the equivalent of omniscient beings That is if there's a God we could become in the far future

or the aliens if we discover them They'll be so far advanced than as they'll appear to be omniscient and omnipotent and that that's the closest we can get to a deity It'd be our future selves or the aliens. All right, dude And Chuck, we still got you on our YouTube channel. Oh, thank you. Just smart enough. Thank you, sir. Chuck, nice. Just smart enough. Look it up and watch it on YouTube, guys. All right. And Michael Sherman delighted to have you. Thank you for having me. This has been another installment of StarTalk Special Edition. Neil deGrasse Tyson, your personal astrophysicist, as always bidding you to keep looking up.